

Automatic Water Tank Controller Using ESP32

Objective

To automatically stop a water pump when the water tank becomes full while allowing the user to manually start the pump using a push button.

Components Used

ESP32 DevKit V1 (1)
IRLZ44N MOSFET (1)
Float Switch (1)
Push Button (1)
Mini Water Pump (5V) (1)
LM2596 Buck Converter (1)
1N4007 Flyback Diode (1)
10 k Ω Resistor (1)
Breadboard (1)
Jumper Wires (As Required)
2 × 18650 Batteries (2)

Working Principle

1. User presses the push button.
2. ESP32 detects button input.
3. ESP32 turns ON the MOSFET.
4. MOSFET powers the pump.
5. Pump fills the tank.
6. Float switch monitors the water level.
7. When full, ESP32 turns OFF the MOSFET.
8. Pump stops and remains OFF until the button is pressed again.

Input Devices

Push Button
Float Switch

Output Device

Mini Water Pump (5V)

Controller

ESP32 DevKit V1

Power Supply

Two 18650 Batteries
LM2596 Buck Converter (5.0 V)

Protection

1N4007 flyback diode across the pump.

Skills Learned

ESP32 Programming, GPIO, INPUT_PULLUP, Debouncing, Latching Logic, MOSFET Control, Buck Converter, Hardware Debugging

Applications

Home Water Tanks, Irrigation Systems, Liquid Filling Systems

Conclusion

The project successfully automates water tank filling while preventing overflow using an ESP32 and MOSFET-based pump control.